

Summary: Performing Quantum Computation with Classical Optics

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Abstract

We demonstrate an implementation of the Variational Quantum Eigensolver using classical optical systems by encoding qubits in photonic polarization and spatial modes. Using wave plates, beam splitters, and dove prisms, we map a UCCSD-inspired VQE circuit for H_2 to experimentally realizable optical operations via Jones-matrix decomposition and read out energies through Stokes-parameter measurements. This optical VQE provides an accessible platform for studying variational algorithms and prototyping photonic circuit implementations.

Introduction

Following O'Malley et al., we begin with the reduced two-qubit electronic Hamiltonian for molecular hydrogen:

$$\hat{H} = g_0\mathbb{I} + g_1Z_0 + g_2Z_1 + g_3Z_0Z_1 + g_4Y_0Y_1 + g_5X_0X_1, \quad (1)$$

where X_i , Y_i , and Z_i are Pauli operators acting on qubit i , and the coefficients g_k depend on molecular geometry (in particular, bond length). This compact form captures the essential electronic structure terms needed for ground-state estimation in a near-term quantum setting.

Estimating the minimum eigenvalue of Eq. ?? is the central task of the Variational Quantum Eigensolver (VQE): prepare a parameterized trial state, measure the expectation value $\langle \hat{H} \rangle$, and iteratively update parameters to approach the ground-state energy. In the standard digital model, this procedure is implemented with a reference state preparation, single-qubit rotations, and entangling gates that realize a UCC-inspired ansatz for H_2 .

This thesis investigates how the same quantum-information workflow can be implemented using classical optics. We map qubit states to photonic degrees of freedom (polarization and spatial mode), and map unitary gate operations to physically realizable optical transformations built from wave plates, beam splitters, and mode-control elements. Jones-matrix and Stokes-parameter formalisms are used to connect the theoretical circuit operations to measurable optical quantities.

The goal is not to claim a universal quantum speedup with classical hardware, but to construct a physically transparent and experimentally accessible platform for studying quantum-circuit structure, variational optimization behavior, and measurement strategies motivated by quantum chemistry. By

grounding the study in the O'Malley Hamiltonian and the H_2 VQE benchmark, this work provides a concrete bridge between quantum-algorithm design and optical laboratory realization.

Preparation of the Quantum Circuit

To implement the VQE algorithm for the Hamiltonian in Eq. ??, we first need to prepare the quantum circuit that corresponds to the UCCSD ansatz for H_2 . This involves decomposing the unitary operator into a sequence of single-qubit rotations and two-qubit entangling gates.

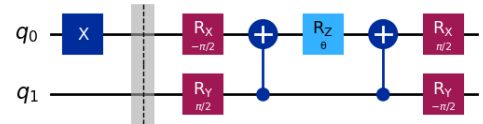


Figure 1: UCCSD ansatz circuit for the hydrogen molecule benchmark used in this work.

Optical implementation of Quantum Gates